Project Initialization and Planning Phase

Date	3 October 2026
Team ID	SWTID-2026-9282
Project Name	ComicCraft - AI Comic Story Creator
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform story creation using generative AI, letting users go from a simple idea toafullyillustratedcomic.Ittacklesthetime,skillandcostbarriersofcomicmaking.Keyfeaturesincludean AI story generator, automatic panel illustration and PDF export.

Project Overview	
Objective	Toletanyoneturnashortideaintoacomplete,illustratedcomicstoryusing generative AI, without needing writing or drawing skills, in a few minutes.
Scope	A web application that takes a story idea, genre and panel count; generates an outline(GeminiFlash),adetailedstorywithpaneldescriptions(GeminiPro)and one image per panel (Stable Diffusion); and exports the finished comic as a PDF.
Problem Statement	
Description	Making a comic needs scripting, illustration and layout skills and usually takes daysofeffort.Students,teachers,writersandhobbyistsoftenhavestoryideasbut cannot turn them into visual stories quickly or affordably.
Impact	Solvingthismakesvisualstorytellingaccessibletoeveryone,cutscreationtime fromdaystominutes,lowerscost,andsupportseducation,contentcreationand creative expression.
Proposed Solution	
Approach	A multi-stage generative AI pipeline: a fast model (Gemini Flash) drafts the outline,astrongermodel(GeminiPro)expandsitintoscenes,dialogueandimage prompts, Stable Diffusion renders each panel, and FPDF assembles everything into a downloadable comic. FastAPI serves the workflow through a simple Jinja2 web interface.
Key Features	- AI-generated story outline from a single prompt (Gemini Flash). - Detailed story and panel-by-panel scripts (Gemini Pro). - Automatic panel image generation (Stable Diffusion via Diffusers). - Comic preview in the browser and one-click PDF export (FPDF).

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPUspecifications, number of cores	T4GPU(16GBVRAM)forStableDiffusion; multi-core CPU
Memory	RAM specifications	16 GB
Storage	Diskspacefordata,models, and logs	50GBSSD(modelweights,generatedimages, PDFs)
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries	Additional libraries	google-generativeai, diffusers, torch, transformers,fpdf,Pillow,python-dotenv
Development Environment	IDE	VS Code, Jupyter Notebook / Google Colab
Data		
Data	Source, size, format	Notrainingdataset;pre-trainedGemini(API) andStableDiffusion(HuggingFace)models. Inputs: text prompts. Outputs: PNG, PDF